

Lecture 13: Validation

Caltech CS156 Technical Study Guide

Gemini Notebook

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Executive Summary

Validation is the primary empirical tool for model selection and hyperparameter tuning in machine learning. While regularization estimates the overfit penalty analytically, validation measures out-of-sample performance directly using unseen data D_{val} . This guide details the mathematical foundations of validation, the trade-offs in choosing the validation set size K , cross-validation techniques, and optimistic selection bias.

1 Validation vs. Regularization

Both regularization and validation serve to control overfitting and estimate out-of-sample performance:

$$E_{\text{out}}(h) \leq E_{\text{in}}(h) + \Omega(\mathcal{H}) \quad (1)$$

- **Regularization:** Minimizes the *augmented error* $E_{\text{aug}}(\mathbf{w}) = E_{\text{in}}(\mathbf{w}) + \lambda\Omega(\mathbf{w})$ analytically by adding a penalty for model complexity.
- **Validation:** Directly estimates E_{out} empirically on an unseen validation set D_{val} of size K . It is used to select optimal hyperparameter values (e.g., λ^*).

2 Mathematical Formulation of Validation

Given a dataset D with N examples, we partition it into a training set D_{train} ($N - K$ examples) and a validation set D_{val} (K examples):

$$E_{\text{val}}(h) = \frac{1}{K} \sum_{\mathbf{x}_k \in D_{\text{val}}} e(h(\mathbf{x}_k), y_k) \quad (2)$$

2.1 Unbiasedness and Variance

For a single validation point $(\mathbf{x}, y)$, the expected validation error is strictly unbiased:

$$\mathbb{E}[E_{\text{val}}(g^-)] = E_{\text{out}}(g^-) \quad (3)$$

where g^- is the hypothesis trained exclusively on D_{train} . Assuming independent sampling:

$$\text{Var}(E_{\text{val}}) = \frac{\sigma^2}{K} \implies E_{\text{out}}(g^-) = E_{\text{val}}(g^-) \pm O\left(\frac{1}{\sqrt{K}}\right) \quad (4)$$

3 The Dilemma of K and Data Restoration

A fundamental trade-off exists regarding K :

- **Small K :** g^- uses $N - K \approx N$ points (so $g^- \approx g$), but E_{val} has high variance and unreliable error bars.
- **Large K :** E_{val} is extremely reliable, but g^- is trained on very few points ($N - K$), making it a poor proxy for the final hypothesis g .

Restoration Rule: After choosing the optimal model/parameter using D_{val} , retrain the chosen model on the *entire* dataset D (N points) to deliver final hypothesis g . **Rule of Thumb:** A standard heuristic is $K = N/5$ (or $N/6$), devoting 20% of the data to validation.

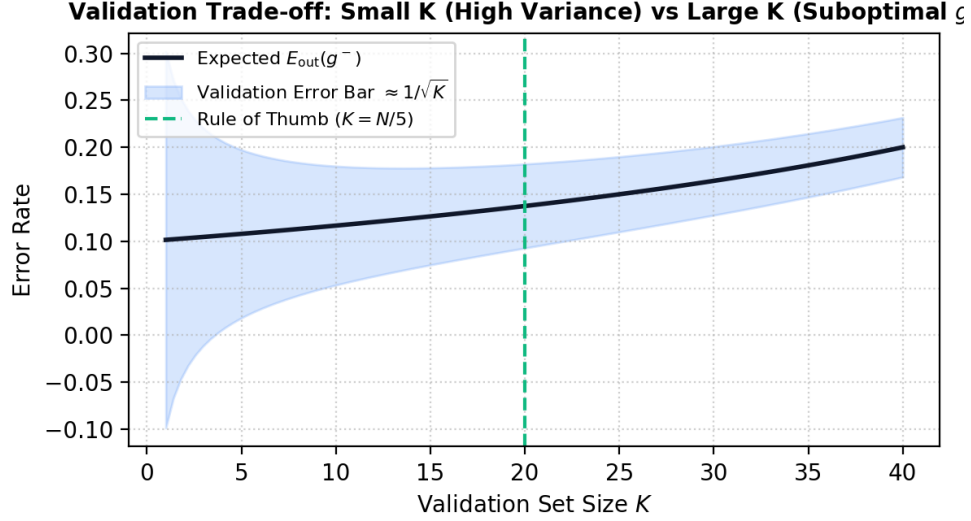


Figure 1: Validation Trade-off: Small K yields high estimation variance; large K leaves insufficient training data ($N - K$), making g^- inferior to g .

4 Model Selection and Optimistic Bias

When validation is used to choose among M models $(H_1, \dots, H_M)$, the minimum validation error $E_{\text{val}}(g_{m^*}^-)$ becomes a **biased, optimistic** estimator:

$$\mathbb{E} \left[\min_m E_{\text{val}}(g_m^-) \right] < \mathbb{E}[E_{\text{out}}(g_{m^*}^-)] \quad (5)$$

The VC/Union bound applies to validation model selection:

$$E_{\text{out}}(g) \leq E_{\text{val}}(g_{m^*}^-) + O \left(\sqrt{\frac{\ln M}{K}} \right) \quad (6)$$

5 Cross-Validation

5.1 Leave-One-Out Cross-Validation (LOOCV)

To maximize training data, set $K = 1$ and repeat N times:

$$E_{\text{cv}} = \frac{1}{N} \sum_{n=1}^N e_n, \quad e_n = e(g_n^-(\mathbf{x}_n), y_n) \quad (7)$$

LOOCV provides an almost unbiased estimate of the expected $E_{\text{out}}(N - 1)$.

5.2 V-Fold Cross-Validation

To reduce computational cost (N training runs in LOOCV), partition D into V equal folds (typically $V = 10$). Train on $V - 1$ folds and validate on the remaining fold, repeating V times:

$$E_{\text{cv}} = \frac{1}{V} \sum_{v=1}^V E_{\text{val}}^{(v)} \quad (8)$$

10-Fold CV is universally recommended in practice as it balances low variance and low computational overhead.